MIL-C-51276(MU)
26 April 1967
(See section 6)

MILITARY SPECIFICATION
CHEMICAL AGENT, TRIPHOSGENE

1. SCOPE

1.1 This specification covers a technical grade of triphosgene (hexachloromethyl carbonate). The formula is $(OCCl_3)_2CO$.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

INTERSTATE COMMERCE COMMISSION

49 CFR 71-90	-	Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

FSC 1365

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(The Interstate Commerce Commission regulations are a part of the Code of Federal Regulations available from the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402. Orders for the above publications should cite "49 CFR 71-90.")

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM STANDARDS

D1193-56 - Reagent Water.

(Application for copies of ASTM Standards should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

3. REQUIREMENTS

3.1 Color and structure. The triphosgene shall be a white crystalline solid, when examined as specified in 4.2.4.1.

3.2 Melting point. The melting point of the triphosgene shall be from 74°C to 79°C inclusive, when tested as specified in 4.2.4.2.

3.3 Chlorine content. The triphosgene shall have a chlorine content between 70.5 and 71.7 percent inclusive, when tested as specified in 4.2.4.3.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of section 5 have been satisfied.

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4.2 Regular production.

4.2.1 Lotting. A lot shall consist of the triphosgene offered for acceptance at the time which has been produced by one manufacturer under essentially the same manufacturing conditions and with no change in materials provided the operation is continuous. In the event that the process is a batch operation, each batch shall constitute a lot (see 6.3).

4.2.2 Sampling. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.3 Examination. Each specimen in the sample shall be tested as specified in 4.2.4. The acceptance number is zero.

4.2.4 Tests. Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.2.4.1 Color and structure. The color and structure of the triphosgene shall be determined visually.

4.2.4.2 Melting point. The melting point of the triphosgene shall be determined by the capillary tube method.

4.2.4.3 Chlorine content. The triphosgene shall be decomposed with sodium peroxide in a Parr bomb and then the percentage of chlorine determined by the Volhard method (see 6.4).

(a) Decomposition. A 0.2 gram sample of triphosgene shall be decomposed with 10 grams of sodium peroxide -- sucrose mix (ratio: 20 to 1) by heating in a Parr bomb. After decomposition is complete, the product shall be washed and filtered.

(b) Determination of chlorine. The decomposed sample solution shall be acidified with 35 percent nitric acid solution plus a 5 milliliter (ml) excess. This solution shall be diluted to approximately 150 ml and 20.0 ml of 0.1 Normal (N) silver nitrate shall be added. After adding 3.0 ml of ferric ammonium sulfate indicator, the solution shall be titrated with 0.1N potassium thiocyanate solution. The percent chlorine shall be calculated using the following formula:

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$$\text{Percent total chlorine} = \frac{[(M_1N_1)-(M_2N_2)]}{W} \times 3.546$$

where:

M_1 = milliliters of silver nitrate,

M_2 = milliliters of potassium thiocyanate,

N_1 = Normality of silver nitrate,

N_2 = Normality of potassium thiocyanate, and

W = weight of sample in grams.

5. PREPARATION FOR DELIVERY

5.1 Cube. Packaging and packing shall be of the minimum cube required to contain the item.

5.2 Packaging, level C. A nominal one pound quantity of triphosgene shall be packaged in a glass jar with screw cap closure to provide adequate protection against corrosion, deterioration and damage from the supply source to the first receiving activity for immediate use.

5.3 Packing, level C. The triphosgene, packaged as specified in 5.2, shall be packed to insure carrier acceptance and safe delivery to the first domestic destination. Containers shall be in compliance with the Interstate Commerce Commission (ICC) regulations for a class A poison, or the regulations of other common carriers applicable to the mode of transportation.

5.4 Marking. In addition to any marking required by the contract or order, unit packages, intermediate packages and shipping containers shall be marked in accordance with the ICC regulations for class A poisons and with MIL-STD-129.

5.4.1 Precautionary marking. All unit packages, intermediate packages and shipping containers shall be marked: Do not allow water to contact contents.

6. NOTES

6.1 Intended use. The triphosgene covered by this specification is intended to be used in the E18 Chemical Agent Identification and Training Set.

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6.2 Ordering data. Procurement documents should specify the title, number and date of this specification.

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Precaution. Triphosgene is a strongly lachrymatory (causing a flow of tears) solid and should be handled with caution and with adequate ventilation. When heated to decomposition, it emits highly toxic fumes of chlorides. When using a Parr bomb, caution should be exercised to prevent explosive rupture. It is suggested that a source of carbon, such as sucrose, be added to the bomb in conjunction with the sample.

6.5 Supersession data. This specification supersedes Edgewood Arsenal Purchase Description 197-54-1017, dated 22 July 1966.

Custodian:

Army - MU

Preparing activity:

Army - MU (EA)

Project No. 1365-A029

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R064

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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